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DIV:SA2
PROJECT TOPIC:E-commerce User Behavior Analysis and Purchase Predicti
DATASET DETAILS
DATASET NAME:E-commerce User Behavior Dataset
SOURCE: PROVIDED DATASET FILE (.xls) FROM KAGGLE
Number of Rows: 8000
Number of Columns: 14
TYPES OF DATA:
The dataset contains different types of data used for analysis:
1. Numerical Data
- Represents measurable quantities.
 - Examples: age, time_on_site, pages_viewed,
 previous_purchases, avg_session_time, bounce_rate
 2. Categorical Data
   - Represents categories or labels.
      - Examples: gender, device_type
      3. Binary Data
   - Contains only two values (0 or 1).
   - Examples: purchase, ad_clicked, returning_user, discount_seen
   4. Identifier Data
   - Unique identifier for each record.
   - Example: user_id
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# E-commerce User Behavior Analysis and Purchase Prediction
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## Problem Statement

#I am analyzing e-commerce user behavior data to understand what factors influence whether a user makes a purchase or not. T
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```
# Import all required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# For model
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

# Settings
sns.set(style="whitegrid")
```

```
# Load dataset
df = pd.read_csv("ecommerce_user_behavior_8000.csv")

# Display first 5 rows
df.head()
### Data Loading and Exploration

#In this section, we load the dataset and explore its structure by viewing the first few rows, dataset shape, and data type
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	user_id	age	gender	device_type	time_on_site	pages_viewed	previous_purchases	cart_items	discount_seen	ad_clicked
0	1.0	56.0	Female	Desktop	12.90	8.0	13.0	1.0	1.0	NaN
1	2.0	46.0	Male	Mobile	15.63	9.0	4.0	6.0	1.0	1.0
2	3.0	32.0	Female	NaN	11.64	12.0	11.0	0.0	0.0	0.0
3	4.0	25.0	Female	Mobile	22.71	5.0	10.0	1.0	0.0	0.0
4	5.0	38.0	Female	Mobile	26.35	9.0	12.0	4.0	1.0	0.0

```
# Shape of dataset
print("Shape:", df.shape)

# Data types
print("\nData Types:\n", df.dtypes)
```

```
Shape: (8000, 14)

Data Types:
 user_id               float64
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age                   float64
gender                 object
device_type            object
time_on_site          float64
pages_viewed          float64
previous_purchases    float64
cart_items            float64
discount_seen         float64
ad_clicked            float64
returning_user        float64
avg_session_time      float64
bounce_rate           float64
purchase              float64
dtype: object
```

```
# Missing values
print(df.isnull().sum())

# Fill missing numeric values with mean
numeric_cols = df.select_dtypes(include=np.number).columns
df[numeric_cols] = df[numeric_cols].fillna(df[numeric_cols].mean())

# Fill categorical with mode
categorical_cols = df.select_dtypes(include='object').columns
for col in categorical_cols:
    df[col].fillna(df[col].mode()[0], inplace=True)
    ### Handling Missing Values

#We checked for missing values in the dataset. Numerical columns were filled using mean values, while categorical columns w
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```
user_id               0
age                   0
gender                0
device_type           0
time_on_site          0
pages_viewed          0
previous_purchases    0
cart_items            0
discount_seen         0
ad_clicked            0
returning_user        0
avg_session_time      0
bounce_rate           0
purchase              0
dtype: int64
/tmp/ipykernel_14932/3945931054.py:11: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df[col].fillna(df[col].mode()[0], inplace=True)
```

```
# Remove duplicates
df.drop_duplicates(inplace=True)

print("After removing duplicates:", df.shape)
## Removing Duplicates

#Duplicate rows were identified and removed to ensure data quality and avoid biased results during analysis.
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After removing duplicates: (8000, 14)
```

```
# Select 2 numeric columns
cols = ['time_on_site', 'pages_viewed']

for col in cols:
    print(f"\nStatistics for {col}")
    print("Mean:", df[col].mean())
    print("Median:", df[col].median())
    print("Mode:", df[col].mode()[0])
    print("Standard Deviation:", df[col].std())
    print("Variance:", df[col].var())
    print("Range:", df[col].max() - df[col].min())
    print("Mid-range:", (df[col].max() + df[col].min()) / 2)
    ## Statistical Analysis

#We calculated statistical measures such as mean, median, mode, standard deviation, variance, range, and mid-range for sele
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Statistics for time_on_site
Mean: 15.714673469387755
Median: 15.714673469387757
Mode: 15.714673469387757
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Standard Deviation: 8.278703243046808
Variance: 68.53692738643372
Range: 29.0
Mid-range: 15.5

Statistics for pages_viewed
Mean: 9.970025510204081
Median: 10.0
Mode: 6.0
Standard Deviation: 5.370949672721792
Variance: 28.847100386910316
Range: 18.0
Mid-range: 10.0
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```
plt.figure()
plt.hist(df['time_on_site'], bins=20)
plt.title("Histogram of Time on Site")
plt.xlabel("Time on Site")
plt.ylabel("Frequency")
plt.show()
#### Histogram Insight

#The histogram shows the distribution of time spent on the website. Most users spend a moderate amount of time, while fewer
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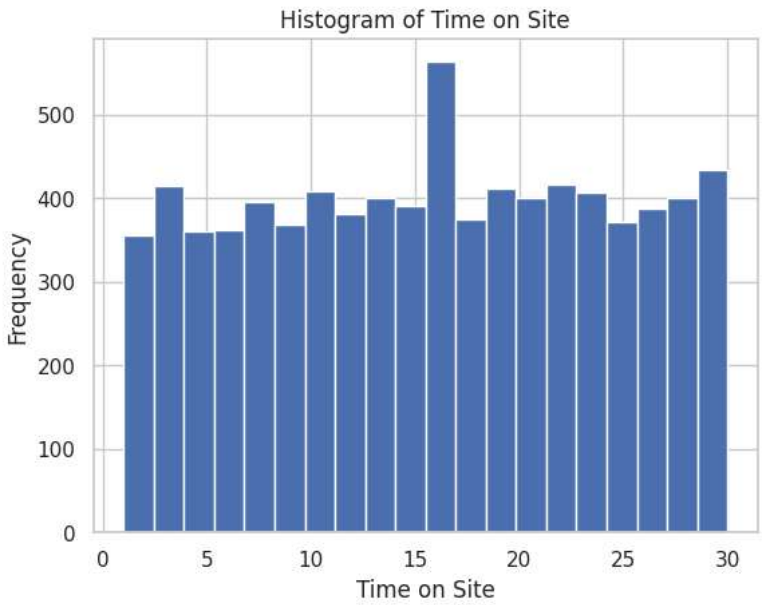
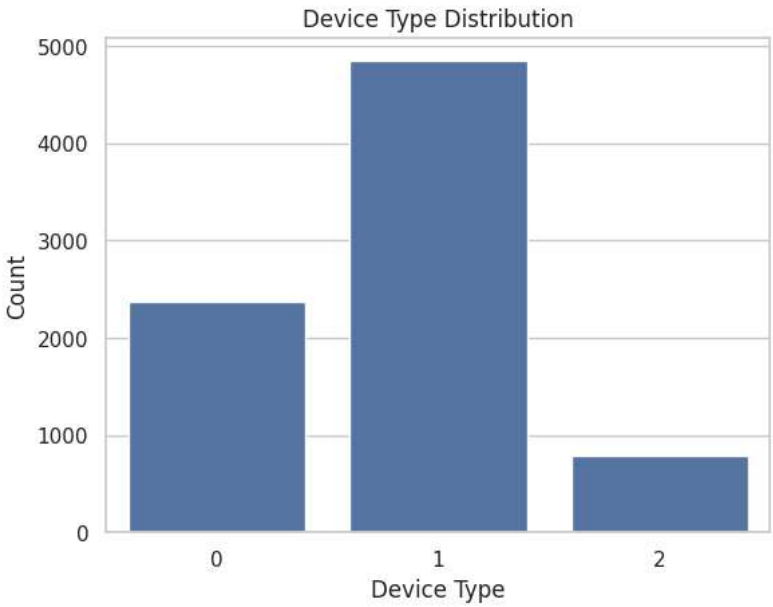



```
plt.figure()
sns.countplot(x='device_type', data=df)
plt.title("Device Type Distribution")
plt.xlabel("Device Type")
plt.ylabel("Count")
plt.show()
```



```
plt.figure()
sns.boxplot(x=df['pages_viewed'])
plt.title("Boxplot of Pages Viewed")
plt.xlabel("Pages Viewed")
plt.show()
#### Boxplot Insight

#The boxplot displays the spread of pages viewed by users and highlights outliers. Some users visit significantly more page
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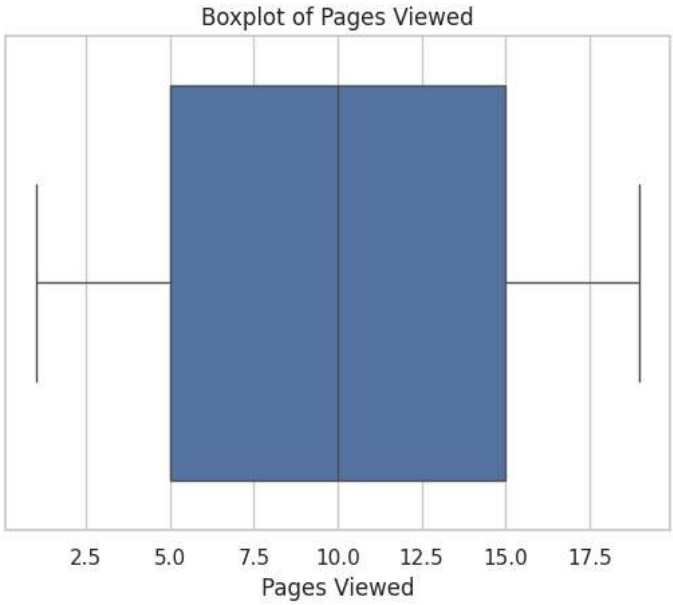


```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
#### Correlation Heatmap Insight

#The heatmap shows relationships between numerical features. Some variables, such as time on site and pages viewed, have a
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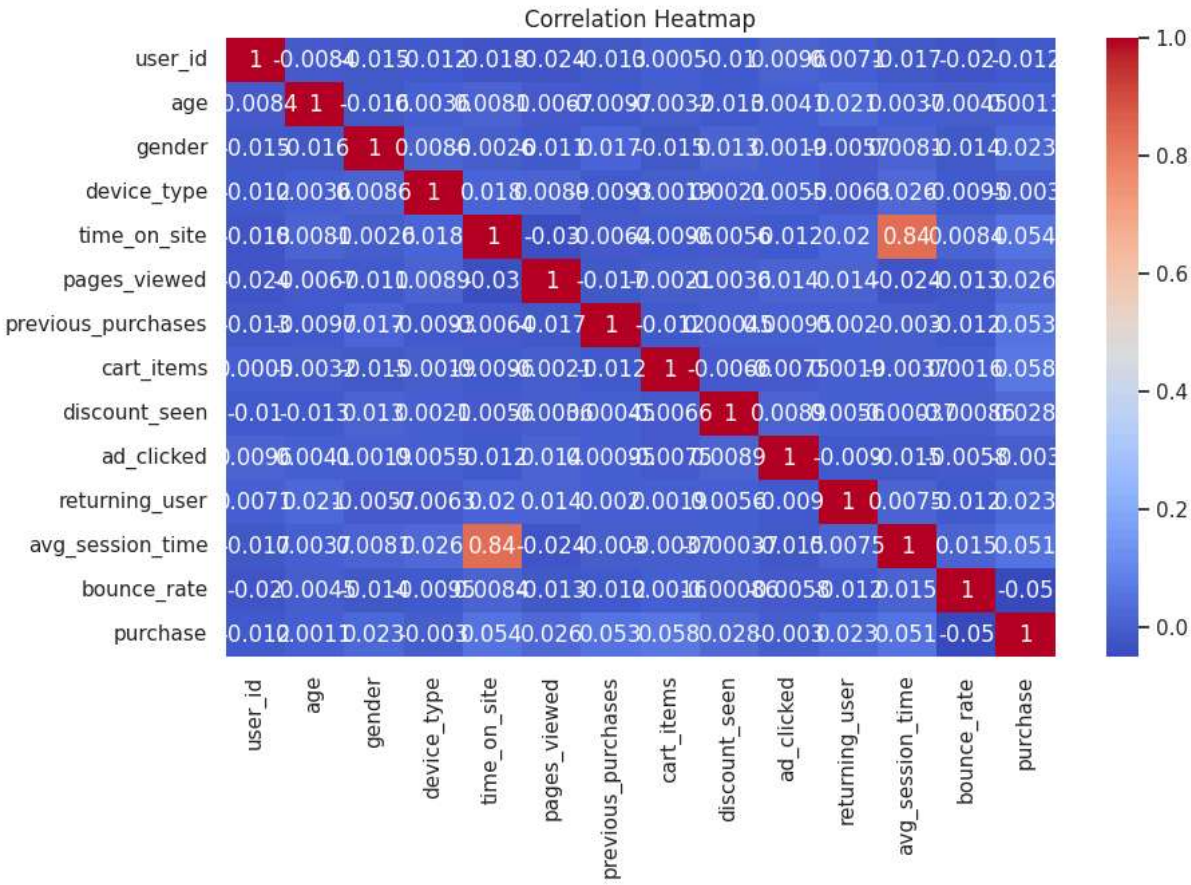


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## Insights
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1. Users who spend more time on the website are more likely to make a purchase, as seen in the histogram.

2. The boxplot shows that users with a higher number of pages viewed tend to have higher engagement, which may lead to purc

3. The correlation heatmap indicates that features like time_on_site and pages_viewed are positively related to purchase be
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Recommendations

The company should focus on increasing user engagement by improving website content and user experience, as higher engagemen

Personalized offers and discounts should be shown to users, especially those who spend more time browsing but do not complet
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